EL1 Worked Examples: Introduction to Exponentials

Recording companion for the EL1 CheckIt support video — Math 172, Santa Ana College

How to use this document

- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

The one question behind every example

What number does the output get multiplied by each time the input goes up by 1? That number is the base b, and the starting value when x is 0 is a. Together they give the exponential function f(x) = a times b to the x.

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Example 1: Table 1

Table 1. Table of x-values and y-values, listed in two rows

x	1	2	3	4	5
y	3	6	12	24	48

Question

What kind of function does this table show?

- **A.** Exponential
- **B.** Linear
- **C.** Neither exponential nor linear

Answer and reasoning

Divide each y-value by the one before it: 6 ÷ 3 = 2, 12 ÷ 6 = 2, 24 ÷ 12 = 2, 48 ÷ 24 = 2. The y-value is multiplied by 2 every time. The differences are 3, 6, 12, 24, so they are not constant. **This table shows an exponential function.**

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Example 2: Table 2

Table 2. Table of x-values and y-values, listed in two rows

x	1	2	3	4	5
y	5	8	11	14	17

Question

What kind of function does this table show?

A. Exponential
B. Linear
C. Neither exponential nor linear

Answer and reasoning

The differences are $8 - 5 = 3$, $11 - 8 = 3$, $14 - 11 = 3$, $17 - 14 = 3$. Adding 3 every time is a constant rate of change, which is linear. The ratios $8 \div 5 = 1.6$ and $11 \div 8 = 1.375$ are not the same. **This table shows a linear function, not an exponential one.**

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Example 3: Table 3

Table 3. Table of x-values and y-values, listed in two rows

x	1	2	3	4	5
y	81	27	9	3	1

Question

What kind of function does this table show?

A. Exponential
B. Linear
C. Neither exponential nor linear

Answer and reasoning

Divide each y-value by the one before it: 27 ÷ 81 = 1/3, 9 ÷ 27 = 1/3, 3 ÷ 9 = 1/3, 1 ÷ 3 = 1/3. The y-value is multiplied by 1/3 every time. The values go down, but a constant ratio is what makes a table exponential. **This table shows an exponential function that decays.**

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Example 4: Table 4

Table 4. Table of x-values and y-values, listed in two rows

x	1	2	3	4	5
y	2	5	10	17	26

Question

What kind of function does this table show?

A. Exponential
B. Linear
C. Neither exponential nor linear

Answer and reasoning

The differences are 3, 5, 7, 9, so they are not constant. The ratios are $5 \div 2 = 2.5$, $10 \div 5 = 2$, $17 \div 10 = 1.7$, $26 \div 17 \approx 1.53$, so they are not constant either. **This table shows neither an exponential nor a linear function.**

Example 5: Equation 1

(0, 5), *(2, 45)*

Question

Which equation describes the exponential function that passes through these two points?

- **A.** $y=3\cdot 5^x$
- **B.** $y=5\cdot 9^x$
- **C.** $y=5\cdot 3^x$
- **D.** $y=20x+5$

Answer and reasoning

The point *(0,5)* has $x=0$, and $b^0=1$, so $a=5$. Now use *(2,45)*: $5\cdot b^2=45$, so $b^2=9$ and $b=3$. The base must be positive. Check: $5\cdot 3^2=45$. **The function is $y=5\cdot 3^x$.**

Example 6: Equation 2

(0, 8), *(3, 1)*

Question

Which equation describes the exponential function that passes through these two points?

- **A.** $y=8\cdot(1/2)^x$
- **B.** $y=8\cdot 2^x$
- **C.** $y=8\cdot(1/8)^x$
- **D.** $y=1/2\cdot 8^x$

Answer and reasoning

The point *(0,8)* gives $a=8$. Now use *(3,1)*: $8\cdot b^3=1$, so $b^3=1/8$ and $b=1/2$. The output goes down from 8 to 1, so a base less than 1 makes sense. Check: $8\cdot(1/2)^3=1$. **The function is $y=8\cdot(1/2)^x$.**

Example 7: Equation 3

$(0, -3)$, $(2, -12)$

Question

Which equation describes the exponential function that passes through these two points?

- **A.** $y=3\cdot 2^x$
- **B.** $y=-3\cdot 4^x$
- **C.** $y=-3\cdot(-2)^x$
- **D.** $y=-3\cdot 2^x$

Answer and reasoning

The point $(0,-3)$ gives $a=-3$. Now use $(2,-12)$: $-3\cdot b^2=-12$, so $b^2=4$. Both 2 and −2 square to 4, but the base of an exponential function must be positive, so $b=2$. Check: $-3\cdot 2^2=-12$. **The function is $y=-3\cdot 2^x$.**

Example 8: Equation 4

(−1, 6), *(0, 2)*

Question

Which equation describes the exponential function that passes through these two points?

- **A.** $y=2\cdot 3^x$
- **B.** $y=2\cdot(1/3)^x$
- **C.** $y=6\cdot(1/3)^x$
- **D.** $y=-4x+2$

Answer and reasoning

This time the point with $x=0$ is the second one, $(0,2)$, so $a=2$. Now use $(-1,6)$: $2\cdot b^{-1}=6$, which means $2/b=6$, so $b=1/3$. Check: $2\cdot(1/3)^{-1}=2\cdot 3=6$. **The function is $y=2\cdot(1/3)^x$.**

Example 9: Evaluate 1

$$f(x)=3\cdot 2^{x}$$

Question

What is the value of $f(4)$?

A. 24
B. 48
C. 1296
D. 12

Answer and reasoning

Substitute 4 for x: $f(4)=3\cdot 2^{4}$. The exponent applies only to the 2, so work out the power first: $2^{4}=16$. Then multiply: $3\cdot 16=48$. Multiplying 3 and 2 first would give $6^{4}=1296$, which is a different problem. ***f(4)=48*.**

Example 10: Evaluate 2

$$g(x)=4\cdot(1/2)^x$$

Question

What is the value of $g(-3)$?

A. $1/8$
B. $1/2$
C. 32
D. 6

Answer and reasoning

Substitute −3 for x: $g(-3)=4\cdot(1/2)^{-3}$. A negative exponent flips the base: $(1/2)^{-3}=2^3=8$. Then multiply: $4\cdot 8=32$. ***g(−3)=32*.**

Example 11: Evaluate 3

$$h(x)=2^{3x-1}$$

Question

What is the value of $h(2)$?

A. 5
B. 10
C. 63
D. 32

Answer and reasoning

The whole expression $3x-1$ is the exponent. Substitute 2 for x: $3(2)-1=5$, so $h(2)=2^5$. Then $2^5=32$. The answer 5 is only the exponent, and 63 comes from subtracting 1 after raising the power. ***h(2)=32*.**

Example 12: Base e 1

$$e^2$$

Question

About how large is this number? Use $e \approx 2.718$.

A. About 2.7
B. About 5.4
C. About 7.4
D. About 9.0

Answer and reasoning

The exponent 2 means two factors of e: $e^2 = e \cdot e \approx 2.718 \cdot 2.718 \approx 7.39$. It is not $2e \approx 5.4$, and it is not $3^2 = 9$, because e is a little less than 3. ***$e^2 \approx 7.4$.***

Example 13: Base e 2

$$f(x)=5e^{x}$$

Question

What is the value of $f(0)$?

A. 0
B. 5
C. $5e$
D. e

Answer and reasoning

Substitute 0 for x: $f(0)=5e^{0}$. Any nonzero number to the power 0 is 1, and that includes e. So $f(0)=5\cdot 1=5$, which is the starting value a. ***f(0)=5*.**

Example 14: Base e 3

$$f(x)=2e^{-x}$$

Question

Does this function show exponential growth or exponential decay?

A. Exponential growth
B. Exponential decay
C. Neither, because the exponent is negative

Answer and reasoning

Rewrite the power using a negative exponent: $e^{-x}=(1/e)^{x}$. The base is $1/e\approx 0.37$, which is between 0 and 1, so the function decays. Seeing the number e in the formula is not enough to call it growth; the sign of the exponent matters. ***f(x)=2e*$^{-x}$ shows exponential decay.**

Example 15: Slip 1

Table 5. Table of x-values and y-values, listed in two rows

x	1	2	3	4
y	2	6	18	54

Question

A student found the differences 4, 12, and 36. They are not equal, so the student says the table is not exponential. What is the mistake, if any?

A. There is no mistake.
B. The differences are equal once you divide each one by 2.
C. Equal differences mean linear; for exponential, check ratios.
D. A table that rises this fast has to be linear, not exponential.

Answer and reasoning

Exponential means multiplying by the same number each step, so divide each y-value by the one before: $6 \div 2 = 3$, $18 \div 6 = 3$, $54 \div 18 = 3$. The ratios are all 3. Unequal differences only rule out linear. **The table is exponential, with $b = 3$.**

Example 16: Slip 2

$$4\cdot b^2=36$$
$$b^2=9$$
$$b=9$$

Question

A student used the points $(0,4)$ and $(2,36)$ and wrote this. What is the mistake, if any?

A. The last step needs a square root, so $b=3$.
B. The 4 should be added, not multiplied, so $b=32$.
C. There is no mistake.
D. The base can be negative here, so $b=-3$.

Answer and reasoning

$b^2=9$ means b times itself is 9, so take the square root: $b=3$. The base of an exponential function must be positive, so -3 is out. Check: $4\cdot 3^2=36$. **The function is $y=4\cdot 3^x$.**

Example 17: Slip 3

$$f(x)=2\cdot 3^x$$
$$f(2)=6^2=36$$

Question

A student wrote this. What is the mistake, if any?

A. There is no mistake.
B. Do the power first: 3 squared is 9, and 2 times 9 is 18.
C. The 2 should be the exponent, so the answer is 9 squared.
D. The answer should be 2 plus 9, which is 11.

Answer and reasoning

The exponent belongs to the 3 only. Find the power first, $3^2=9$, then multiply: $2\cdot 9=18$. Multiplying 2 and 3 first changes the problem to 6^2. **The correct value is $f(2)=18$.**

Example 18: Slip 4

$$f(x)=3e^x$$
$$f(0)=3e^0=3$$

Question

A student wrote this. What is the mistake, if any?

A. The answer should be 3 times e, about 8.15.
B. The answer should be 0, since anything times 0 is 0.
C. There is no mistake.
D. The answer should be 1, since e to the 0 is 1.

Answer and reasoning

Any nonzero number to the 0 power is 1, and e is a number, so $e^0=1$. Then $3\cdot 1=3$. The value at $x=0$ is always the starting value a. **There is no mistake: $f(0)=3$.**

Example 19: Slip 5

$$5\cdot(1/2)^{-2}=-5/4$$

Question

A student wrote this. What is the mistake, if any?

A. The answer should be 5 times one fourth, which is positive.
B. There is no mistake.
C. The answer should be negative 10, from 5 times negative 2.
D. Negative exponents flip the base, so the value is 20.

Answer and reasoning

A negative exponent means flip the base, not make the answer negative: $(1/2)^{-2}=2^2=4$. Then $5\cdot 4=20$. An exponential function with a positive a never gives a negative value. **The correct value is 20.**